

|     |          |                                                                                                                                                                                                                                                                                |
|-----|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |          | $I_1 = I_3 \left( \frac{R_3}{R_2} + 1 \right)$ , giving $\frac{I_3}{I_1} = \frac{R_2}{R_2 + R_3}$                                                                                                                                                                              |
| 22. | <b>C</b> | $B = \frac{\mu_0 I}{2\pi d}$ <p>Since we want to find the distance at which the PEAK flux density would be, that would occur when the PEAK current is flowing through the circuit.</p> $100 \times 10^{-6} = \frac{\mu_0 (2000 \times \sqrt{2})}{2\pi d}$ $d = 5.66 \text{ m}$ |
| 23. | <b>B</b> | Current carrying conductors will attract when currents are in same direction                                                                                                                                                                                                   |